M.Tech. Semester – III (Mid-Semester) Examination, 2025

Subject: Computer Science and Engineering
Paper Code & Name: CSE-1103: Elective V (VLSI Design)
Full Marks: 30

Date: 08.12.2025 **Time and Duration: 12:30 PM – 02:00 PM; 90 Minutes**

<u>You are informed to note down the following instructions carefully:</u>
Promise not to commit any academic dishonesty.
Candidates are required to answer in their own words as far as applicable.

Answer <u>any four</u> from the following questions:

1. Write short notes on—

(a) Size and complexity of integrated circuits. [3½]

(b) Benefits of increase in device count. [4]

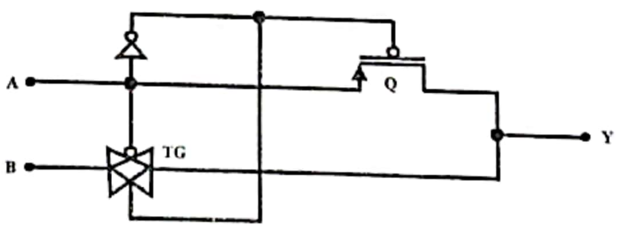
2. Draw the CMOS inverter circuit and show how it operates. State the structural and functional differences between CMOS and nMOS inverter circuits. [4+3½]

3. Implement the following function using CMOS logic—

$$f = \Sigma_m (1, 4, 5, 6, 7, 11, 12, 13, 14, 15).$$ [7½]

4. (a) "The propagation delay of a 3-input CMOS NOR gate is more than that of a 3-input CMOS OR gate."— Justify with necessary diagram(s). [3½]

(b) Identify the following circuit— [4]



5. (a) Briefly describe how a transmission gate operates when the control input is high. [3½]

(b) Show that at most two transmission gates are sufficient to realize any 2-variable function in the form of CMOS combinational network. Give examples. [4]